EXP 3

```
clc;
clear;
close all;

% Processing Gain and PN Sequence Generation
% for Spread Spectrum Communication

% Step 1: Define parameters
N = 7;                % Number of PN sequence bits
dataBits = [1 0 1 1 0 1 0];

% Step 2: Generate PN sequence using a 3-bit LFSR
% Feedback polynomial: x^3 + x + 1
register = [1 1 1];
pn = zeros(1, 2^N - 1);

for k = 1:length(pn)
   pn(k) = register(end);
   feedback = xor(register(3), register(1));
   register = [feedback register(1:2)];
end

% Use a 7-chip PN sequence
pn = pn(1:7);

% Step 3: Calculate processing gain
Rb = 1e3;             % Data rate in bits/sec
Rc = 7e3;             % Chip rate in chips/sec
PG = Rc / Rb;         % Processing gain
PG_dB = 10*log10(PG);   % Processing gain in dB
```

```
% Step 4: Spread the data
spreadData = zeros(1, length(dataBits)*length(pn));

for k = 1:length(dataBits)
    chips = xor(dataBits(k), pn);
    idx = (k-1)*length(pn) + (1:length(pn));
    spreadData(idx) = chips;
end

% Convert binary chips to bipolar form
pn_bipolar = 1 - 2*pn;
data_bipolar = 1 - 2*dataBits;
spread_bipolar = 1 - 2*spreadData;

% Step 5: Display results
disp('Generated PN Sequence:');
disp(pn);

fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain = %.2f dB\n', PG_dB);

disp('Original Data Bits:');
disp(dataBits);

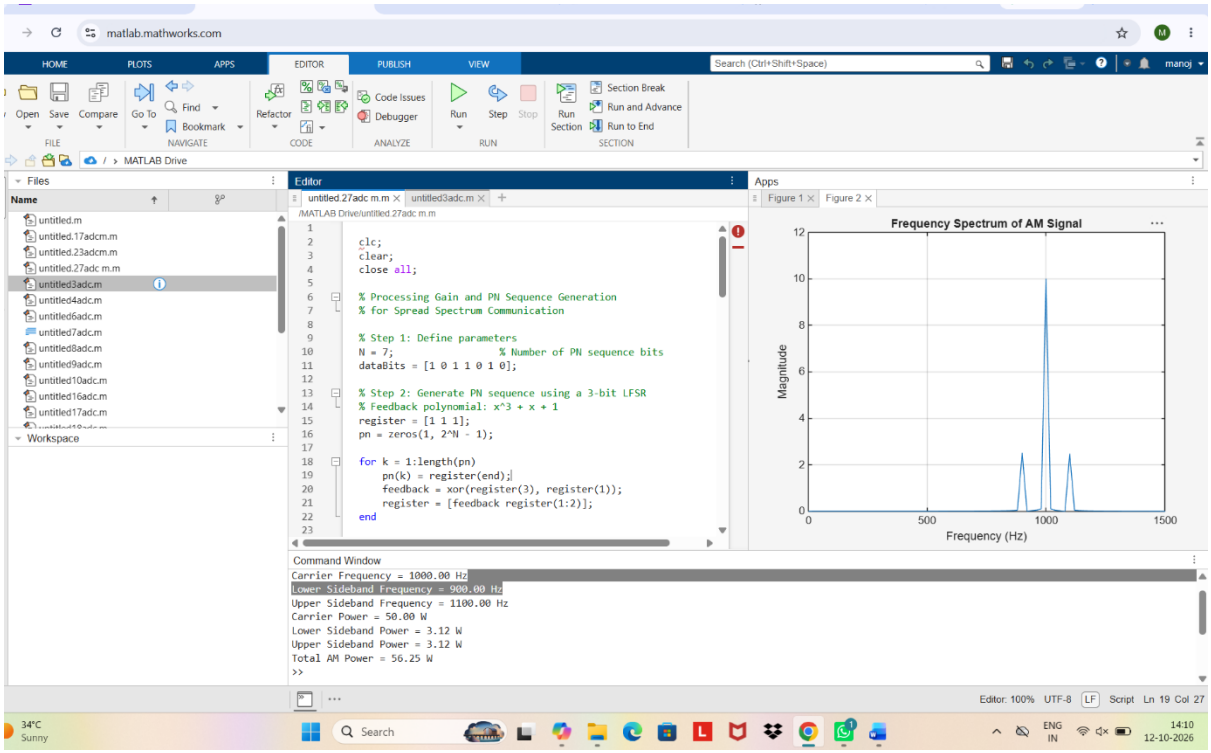
disp('Spread Data Bits:');
disp(spreadData);

% Step 6: Plot results
figure;
```

```
subplot(3,1,1);
stairs(0:length(dataBits)-1, dataBits, 'LineWidth', 1.5);
title('Original Data Bits');
xlabel('Bit Index');
ylabel('Bit Value');
grid on;
ylim([-0.2 1.2]);

subplot(3,1,2);
stairs(0:length(pn)-1, pn, 'LineWidth', 1.5);
title('Generated PN Sequence');
xlabel('Chip Index');
ylabel('Chip Value');
grid on;
ylim([-0.2 1.2]);

subplot(3,1,3);
stairs(0:length(spreadData)-1, spreadData, 'LineWidth', 1.5);
title('Spread Spectrum Data');
xlabel('Chip Index');
ylabel('Chip Value');
grid on;
ylim([-0.2 1.2]);
```


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feedback = xor(register(3), register(1));
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end
Apps
Figure 1
Figure 2
Frequency Spectrum of AM Signal
Magnitude
Frequency (Hz)
0
2
4
6
8
10
12
0
500
1000
1500
Command Window
Carrier Frequency = 1000.00 Hz
Lower Sideband Frequency = 900.00 Hz
Upper Sideband Frequency = 1100.00 Hz
Carrier Power = 50.00 W
Lower Sideband Power = 3.12 W
Upper Sideband Power = 3.12 W
Total AM Power = 56.25 W
>>
Editor: 100%
UTF-8
LF
Script
Ln 19 Col 27
34°C
Sunny
Search
ENG
IN
14:10
12-10-2026